Main Code (`cloud_storage_tier.py`)

```
class CloudStorageTier:
    def __init__(self):
        self.storage_data = {}

    def upload_file(self, file_name, file_size, tier="standard"):
        if not file_name:
            return "Error: Invalid file name"
        if file_size < 0:
            return "Error: Invalid file size"

        # Determine tier logic
        if tier == "archive" and file_size > 100:
            cost = file_size * 0.01
        elif tier == "standard":
            cost = file_size * 0.05
        else:
            cost = file_size * 0.03

        self.storage_data[file_name] = {"size": file_size, "tier": tier, "cost": cost}
        return f"File {file_name} uploaded successfully to {tier} tier."

    def get_file_info(self, file_name):
        if file_name in self.storage_data:
            return self.storage_data[file_name]
        return "Error: File not found"

    def delete_file(self, file_name):
        if file_name in self.storage_data:
```

```
            del self.storage_data[file_name]
            return f"File {file_name} deleted successfully."
        return "Error: File not found"
```

2) Test Code (`test_cloud_storage.py`)

```
import pytest
from cloud_storage_tier import CloudStorageTier

def test_upload_success():
    storage = CloudStorageTier()
    result = storage.upload_file("test1.txt", 50, "standard")
    assert "uploaded successfully" in result

def test_upload_invalid_filename():
    storage = CloudStorageTier()
    result = storage.upload_file("", 50, "standard")
    assert result == "Error: Invalid file name"

def test_upload_invalid_size():
    storage = CloudStorageTier()
    result = storage.upload_file("test2.txt", -10, "standard")
    assert result == "Error: Invalid file size"

def test_upload_archive_tier():
    storage = CloudStorageTier()
    result = storage.upload_file("archive_file.zip", 150, "archive")
    assert "uploaded successfully" in result
```

```
def test_upload_other_tier():
    storage = CloudStorageTier()
    result = storage.upload_file("cold_file.zip", 50, "cold")
    assert "uploaded successfully" in result

def test_get_file_info():
    storage = CloudStorageTier()
    storage.upload_file("info.txt", 20, "standard")
    info = storage.get_file_info("info.txt")
    assert info["size"] == 20

def test_get_file_not_found():
    storage = CloudStorageTier()
    result = storage.get_file_info("nonexistent.txt")
    assert result == "Error: File not found"

def test_delete_file():
    storage = CloudStorageTier()
    storage.upload_file("del.txt", 10, "standard")
    result = storage.delete_file("del.txt")
    assert "deleted successfully" in result

def test_delete_file_not_found():
    storage = CloudStorageTier()
    result = storage.delete_file("nonexistent.txt")
    assert result == "Error: File not found"
```